

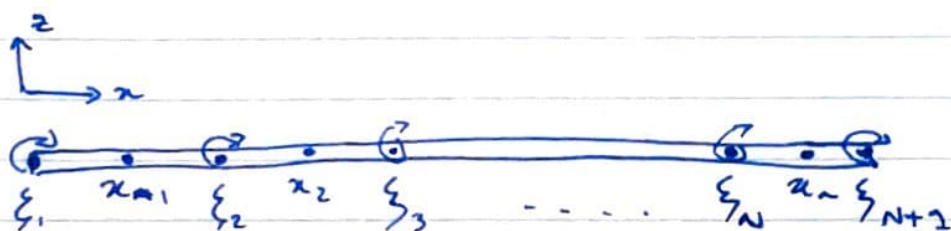
Calculating $V_n^{(k)}$ from one time variable.

We want to express our gust function, $V(t)$, as dependent on time. The simplest way to do this is to express the gust strength V as a function of a single time variable t .

→ This makes it a lot easier to generalise all sorts of gust shapes when implementing into our code.

However, how do we capture the gradual propagation of the gust over our airfoil?

Consider our airfoil as a flat plate like so:



Where ξ_n denotes the position of vortex n , and x_n denotes the position of control point x . Note that x_n lies at the midpoint between ξ_n and ξ_{n+1} .

Since we are calculating Γ using the exact distances x_n , it would be wise to also calculate V at x_n . Lysak demonstrates examples where he considers the whole panel:

$$V_n^{(k)} = V_0 \Gamma(k-n), \Gamma = \text{heaviside function}$$

This is ok for discrete-related functions such as the heaviside function, but fails in accuracy for more continuous ones.

Thus, we need a way to:

- Express V as a function of one time variable t , and
- make $V(t)$ evaluate precisely at x_n .

Solution

Let us note that for any time step k , we can find the total time passed as:

$$t = \Delta t(k-1)$$

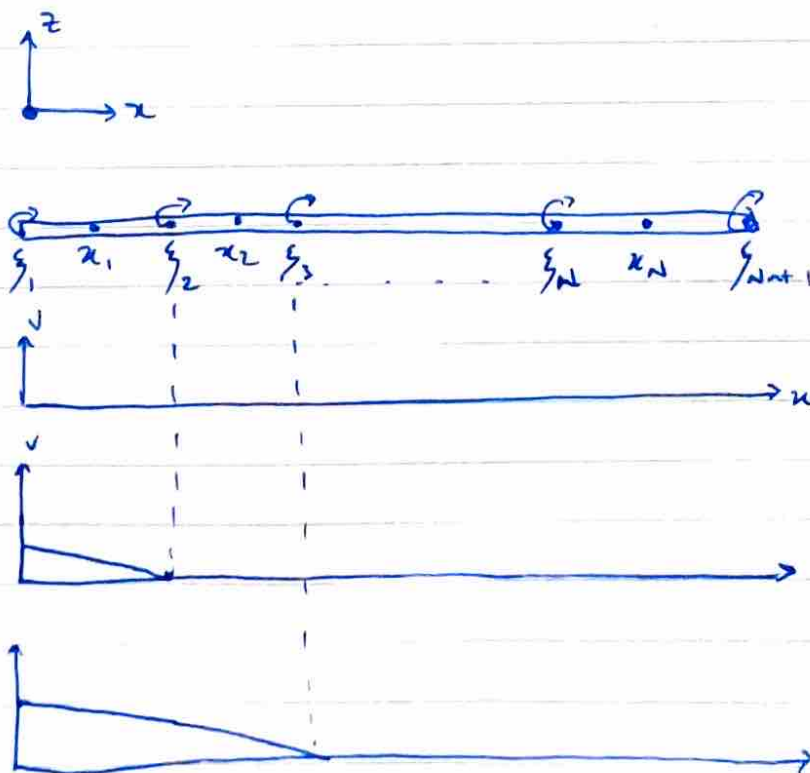
Where Δt is the time step length, given by $\Delta t = \Delta x / U_\infty$. Note that this represents the time it takes for the freestream to move one panel length. Also note that $(k-1)$ is used. This is because $k=1$ denotes time step 1 as the initial condition, i.e.

$$t = 0 \text{ at } k=1.$$

Let us consider an example of a ramp gust. The gust shape may take the form

$$V(t) = V_0 t H(t)$$

Note, we are using the Heaviside function to represent a gust starting from 0 at $t=0$. This gust strength will propagate down the airfoil as follows:



To account for the fact that the value for v at some distance x_n along the airfoil, [we note that] v is effectively 'delayed' by a certain amount of time, this being:

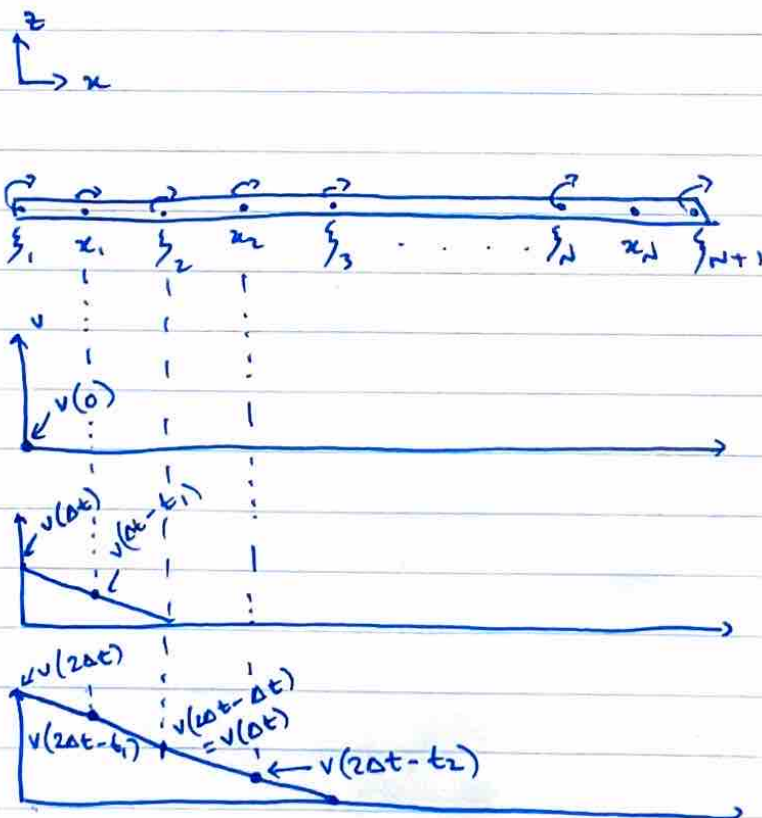
$$t_n = x_n / U_\infty$$

Thus, given that x_n is the distance to control point n along the chord, we can account for this delay in our final expression for v ; by evaluating v at $t - t_n$, and defining t as the time since the start of the gust function (the zero point) hits the leading edge, we can accurately evaluate v at control point n using a single parameter.

That is, let $v(t)$ denote the gust strength at time t , where t is the time since the gust behaviour hits the leading edge. Then;

$$v_n^{(k)} = v(t - t_n), \text{ where } t = \Delta t(k-1), t_n = \frac{x_n}{U_\infty}.$$

Visually:



Thus, we can now implement time-stepping logic whilst generalising our function v .